

Sami Hatoum

07741 157779 | s.hatoum@proton.me | linkedin.com/in/sami-hatoum-dev | github.com/Sami-Hat | www.samihatoum.dev

Education

University College London | *MSc Artificial Intelligence and Imaging* 2025 – 2026

Predicted Distinction

Dissertation: *Fair Uncertainty Quantification for Selective Prediction in Medical Image Segmentation*

Modules: Applied Deep Learning, Machine Learning in Medical Imaging, Statistical Modelling, Inverse Problems

University of St Andrews | *BSc Computer Science* 2021 – 2025

First Class Honours, Dean's List, Honours Project Distinction

Dissertation: *Sokoban Procedural Instance Generation* (constraint optimisation over a PSPACE-complete design space)

Modules: Computational Complexity, Constraint Programming, Data Structures and Algorithms, Databases

Skills

Languages: Python, Java, JavaScript, TypeScript, C++, SQL

Web: React, Next.js, Node.js, Express.js, Flask, FastAPI, REST APIs

Machine Learning: PyTorch, scikit-learn, Hugging Face Transformers, MONAI, Pandas, NumPy

Cloud & DevOps: AWS (Certified Cloud Practitioner), Docker, Git, GitHub Actions, CI/CD, Linux

Databases & Testing: PostgreSQL, Redis, Jest, Pytest, Locust, Playwright

Experience

PricewaterhouseCoopers (PwC) | *Software Engineering Intern* Jun 2024 – Aug 2024

- Partnered with **ML engineers** to build a **PyTorch BERT** sentiment model automating daily classification of **10,000 customer reviews** for a retail client; achieved **91% F1** on an imbalanced 3:1 negative-to-positive dataset
- Deployed the model as a containerised **Flask REST API** on **AWS EC2** with auto-scaling; validated **sub-500ms latency** at **150 requests/second** under load testing with **Locust**
- Built **CloudWatch** dashboards tracking F1 drift; alert thresholds informed rollback decisions after accuracy regression
- Optimised preprocessing with **pandas** vectorisation, reducing daily batch processing time by **60%**

PricewaterhouseCoopers (PwC) | *Software Engineering Intern* Aug 2023 – Sep 2023

- Developed an internal **client onboarding dashboard** in **React** and **Node.js** replacing a manual spreadsheet workflow for a financial services team, reducing onboarding time from **3 days to 4 hours** across 200+ users
- Designed a **PostgreSQL** data model with **role-based access control** across 4 user roles, working with a data scientist on schema requirements; integrated with PwC's internal SSO via **OAuth 2.0**
- Wrote **CI/CD pipelines** in **GitHub Actions** with linting, automated test gates, and staged deployments to **AWS**; zero failed releases after pipeline adoption
- Achieved **90%+ code coverage** across **45 unit and integration tests** (**Jest**, **Supertest**) spanning the REST API and data access layers

Projects

Real-Time Collaborative Code Editor

- Built a multi-user code editor in **React** and **Node.js** using **WebSockets** and **Operational Transform** for conflict-free concurrent editing, with syntax highlighting for 12 languages via **CodeMirror**
- Implemented a **Redis** pub/sub layer for horizontal scaling across server instances, sustaining **50+ concurrent editors** at **sub-100ms synchronisation latency** under benchmark, with automatic reconnection on network loss
- Added **JWT authentication**, per-room permission levels (owner/editor/viewer), and a session persistence layer in **PostgreSQL** enabling users to rejoin documents with full edit history intact

Database Query Optimiser

- Developed a **PostgreSQL** analysis tool in **Python** parsing query ASTs and evaluating **EXPLAIN plans** to recommend indexes; reduced execution time from **2.4s to 420ms** on table scans over a synthetic 500K-row dataset
- Analysed **200+ generated queries** across a synthetic **five-million-row schema** (12 tables), identifying **21 missing indexes** that reduced average query time by **65%** and eliminated most sequential scans. Dockerised for reproducibility

Fair Uncertainty Quantification for Selective Prediction in Medical Image Segmentation (MSc Dissertation)

- Investigated calibration and distributional fairness of segmentation uncertainty estimates, building a selective prediction framework deferring unreliable outputs to clinical review rather than failing silently
- Achieved **0.89 Dice at 85% coverage** on 9-class pelvic MRI segmentation (**+0.02 Dice** over full-coverage baseline) using **nnU-Net** with **Monte Carlo Dropout**, comparing entropy and mutual information for uncertainty aggregation

Leadership

St Andrews AI Society | *Machine Learning Officer* Sep 2024 – Apr 2025

- Led weekly interactive workshops on machine learning for **50+ members** spanning technical and non-technical backgrounds, designing sessions tractable across a wide range of mathematical and programming experience